

SPRAWOZDANIE

PROGRAMOWANIE SKRYPTOWE – LABORATORIUM 1

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Zad.1

a) ls -la

```
(kali㉿kali)-[~]
└─$ ls -la
total 4108
drwx----- 22 kali kali    4096 Mar 12 04:08 .
drwxr-xr-x  3 root root    4096 Sep  9  2025 ..
-rw-r--r--  1 kali kali     220 Sep  9  2025 .bash_logout
-rw-r--r--  1 kali kali    5551 Sep  9  2025 .bashrc
-rw-r--r--  1 kali kali    3526 Sep  9  2025 .bashrc.original
drwx-----  8 kali kali    4096 Nov 26 13:59 .BurpSuite
drwxrwxr-x 11 kali kali    4096 Nov 26 14:10 .cache
drwxr-xr-x 15 kali kali    4096 Nov 26 14:10 .config
drwxr-xr-x  2 kali kali    4096 Mar 11 14:40 Desktop
-rw-r--r--  1 kali kali     35 Nov 25 16:02 .dmrc
drwxr-xr-x  2 kali kali    4096 Dec  3 09:29 Documents
drwxr-xr-x  2 kali kali    4096 Nov 26 14:10 Downloads
-rw-r--r--  1 kali kali   11759 Sep  9  2025 .face
lrwxrwxrwx  1 kali kali      5 Sep  9  2025 .face.icon → .face
drwx-----  3 kali kali    4096 Nov 25 16:02 .gnupg
-rw-----  1 kali kali      0 Nov 25 16:02 .ICEauthority
drwxr-xr-x  4 kali kali    4096 Nov 26 13:31 .java
drwx-----  2 kali kali    4096 Mar 12 03:13 .john
drwxrwxr-x  4 kali kali    4096 Mar 11 15:18 katalog2
drwxr-xr-x  5 kali kali    4096 Nov 25 16:02 .local
-rw-rw-rw-  1 kali kali   1592036 Dec  3 16:16 logs.txt
drwx-----  4 kali kali    4096 Nov 25 16:02 .mozilla
drwxr-xr-x  2 kali kali    4096 Nov 25 16:02 Music
-rw-rw-r--  1 kali kali   1194027 Dec  3 16:30 odszyfrowane.jpg
-rw-rw-r--  1 kali kali   1194027 Dec  3 16:30 odszyfrowane.txt
drwxr-xr-x  2 kali kali    4096 Nov 25 16:02 Pictures
drwx-----  3 kali kali    4096 Nov 26 13:42 .pki
-rw-rw-r--  1 kali kali     123 Dec  3 10:44 plik.txt
-rw-r--r--  1 kali kali     807 Sep  9  2025 .profile
drwxr-xr-x  2 kali kali    4096 Nov 25 16:02 Public
drwx-----  2 kali kali    4096 Dec  3 10:33 .ssh
-rw-r--r--  1 kali kali      0 Nov 25 16:03 .sudo_as_admin_successful
drwxr-xr-x  2 kali kali    4096 Nov 25 16:02 Templates
drwxrwxr-x  4 kali kali    4096 Mar 11 15:32 test
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-clipboard-tty7-control.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-clipboard-tty7-service.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-display-svgx-x11-tty7-control.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-display-svgx-x11-tty7-service.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-draganddrop-tty7-control.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-draganddrop-tty7-service.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-hostversion-tty7-control.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-seamless-tty7-control.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-seamless-tty7-service.pid
-rw-r----- 1 kali kali      5 Mar 12 02:54 .vboxclient-vmxvga-session-tty7-control.pid
drwxr-xr-x  2 kali kali    4096 Nov 25 16:02 Videos
-rw-----  1 kali kali      49 Mar 12 02:54 .Xauthority
-rw-----  1 kali kali   12222 Mar 12 04:08 .xsession-errors
-rw-----  1 kali kali    9536 Mar 11 14:50 .xsession-errors.old
-rw-r--r--  1 kali kali     336 Sep  9  2025 .zprofile
-rw-----  1 kali kali    2802 Mar 12 04:08 .zsh_history
-rw-r--r--  1 kali kali   10855 Sep  9  2025 .zshrc
```

b) touch plik, ls -l, plik posiada odczyt i zapis na wlasciciela, odczyt i zapis dla grupy i odczyt dla reszty

```
(kali㉿kali)-[~]
$ touch plik

(kali㉿kali)-[~]
$ ls -l
total 3936
drwxr-xr-x 2 kali kali 4096 Mar 11 14:40 Desktop
drwxr-xr-x 2 kali kali 4096 Dec 3 09:29 Documents
drwxr-xr-x 2 kali kali 4096 Nov 26 14:10 Downloads
drwxrwxr-x 4 kali kali 4096 Mar 11 15:18 katalog2
-rw-rw-rw- 1 kali kali 1592036 Dec 3 16:16 logs.txt
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Music
-rw-rw-r-- 1 kali kali 1194027 Dec 3 16:30 odszyfrowane.jpg
-rw-rw-r-- 1 kali kali 1194027 Dec 3 16:30 odszyfrowane.txt
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Pictures
-rw-rw-r-- 1 kali kali 0 Mar 12 04:10 plik
-rw-rw-r-- 1 kali kali 123 Dec 3 10:44 plik.txt
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Public
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Templates
drwxrwxr-x 4 kali kali 4096 Mar 11 15:32 test
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Videos
```

c)mkdir katalog

```
(kali㉿kali)-[~]
$ mkdir katalog
```

d) cp plik katalog

```
(kali㉿kali)-[~]
$ cp plik katalog

(kali㉿kali)-[~]
$ ls -l
total 3940
drwxr-xr-x 2 kali kali 4096 Mar 11 14:40 Desktop
drwxr-xr-x 2 kali kali 4096 Dec 3 09:29 Documents
drwxr-xr-x 2 kali kali 4096 Nov 26 14:10 Downloads
drwxrwxr-x 2 kali kali 4096 Mar 12 04:12 katalog
drwxrwxr-x 4 kali kali 4096 Mar 11 15:18 katalog2
-rw-rw-rw- 1 kali kali 1592036 Dec 3 16:16 logs.txt
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Music
-rw-rw-r-- 1 kali kali 1194027 Dec 3 16:30 odszyfrowane.jpg
-rw-rw-r-- 1 kali kali 1194027 Dec 3 16:30 odszyfrowane.txt
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Pictures
-rw-rw-r-- 1 kali kali 0 Mar 12 04:10 plik
-rw-rw-r-- 1 kali kali 123 Dec 3 10:44 plik.txt
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Public
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Templates
drwxrwxr-x 4 kali kali 4096 Mar 11 15:32 test
drwxr-xr-x 2 kali kali 4096 Nov 25 16:02 Videos

(kali㉿kali)-[~]
$ cd katalog

(kali㉿kali)-[~/katalog]
$ ls
plik
```

e) rm plik

```
(kali㉿kali)-[~]
$ rm plik

(kali㉿kali)-[~]
$ ls
Desktop  Downloads  katalog2  Music  odszyfrowane.txt  Public  test
Documents katalog  logs.txt  odszyfrowane.jpg  Pictures  Templates Videos
```

f) cd katalog

```
(kali㉿kali)-[~]
$ cd katalog

(kali㉿kali)-[~/katalog]
$
```

g) pwd

```
(kali㉿kali)-[~/katalog]
$ pwd
/home/kali/katalog
```

h) chmod 700 plik

```
(kali㉿kali)-[~/katalog]
$ chmod 700 plik

(kali㉿kali)-[~/katalog]
$ ls -l
total 0
-rwx----- 1 kali kali 0 Mar 12 04:12 plik
```

i) touch plik2, chmod 060 plik2

```
(kali㉿kali)-[~/katalog]
$ touch plik2

(kali㉿kali)-[~/katalog]
$ chmod 060 plik2

(kali㉿kali)-[~/katalog]
$ ls -l
total 0
-rwx----- 1 kali kali 0 Mar 12 04:12 plik
-----rw 1 kali kali 0 Mar 12 04:16 plik2
```

j) chmod -R 777 .

```
(kali㉿kali)-[~/katalog]
$ chmod -R 777 .

(kali㉿kali)-[~/katalog]
$ ls -l
total 0
-rwxrwxrwx 1 kali kali 0 Mar 12 04:12 plik
-rwxrwxrwx 1 kali kali 0 Mar 12 04:16 plik2
```

k) mv plik zadanie1

```
(kali㉿kali)-[~/katalog]
$ mv plik zadanie1

(kali㉿kali)-[~/katalog]
$ ls
plik2 zadanie1
```

Zad.2

a) find ~ -name zadanie1

```
(kali㉿kali)-[~]  
$ find ~ -name zadanie1  
/home/kali/katalog/zadanie1
```

locate zadanie1

```
(kali㉿kali)-[~]  
$ locate zadanie1  
/home/kali/katalog/zadanie1
```

b) find /etc -name „*.txt”

```
(kali㉿kali)-[~]  
$ find /etc -name "*.txt"  
/etc/theHarvester/wordlists/names_small.txt  
/etc/theHarvester/wordlists/api_endpoints.txt  
/etc/theHarvester/wordlists/general/common.txt  
/etc/theHarvester/wordlists/dns-names.txt  
/etc/theHarvester/wordlists/dorks.txt  
/etc/theHarvester/wordlists/dns-big.txt  
find: '/etc/ipsec.d/private': Permission denied  
find: '/etc/openvas/gnupg': Permission denied  
find: '/etc/credstore': Permission denied  
find: '/etc/ssl/private': Permission denied  
/etc/X11/rgb.txt  
find: '/etc/vpnc': Permission denied  
find: '/etc/polkit-1/rules.d': Permission denied  
/etc/java-21-openjdk/security/policy/README.txt  
find: '/etc/credstore.encrypted': Permission denied  
/etc/unicornscaan/ports.txt  
/etc/unicornscaan/oui.txt  
find: '/etc/redis': Permission denied  
/etc/arp-scan/mac-vendor.txt
```

c) find ~ -type f -perm 777

```
(kali㉿kali)-[~]  
$ find ~ -type f -perm 777  
/home/kali/katalog/plik2  
/home/kali/katalog/zadanie1
```

Zad.3

a) wc -w < /etc/passwd

```
(kali㉿kali)-[~]
$ wc -w < /etc/passwd
96
```

b) `ping -c 5 127.0.0.1 > logi.txt`

```
(kali㉿kali)-[~]
$ ping -c 5 127.0.0.1 > logi.txt

(kali㉿kali)-[~]
$ ls
Desktop  Documents  Downloads  katalog  katalog2  logi.txt  logs.txt

(kali㉿kali)-[~]
$ cat logi.txt
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.285 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.133 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.075 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.107 ms

— 127.0.0.1 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4077ms
rtt min/avg/max/mdev = 0.057/0.131/0.285/0.081 ms
```

c) `cat /etc/passwd | grep -w root > ~/katalog/zadanie1`

```
(kali㉿kali)-[~/katalog]
$ cat /etc/passwd | grep -w root > ~/katalog/zadanie1

(kali㉿kali)-[~/katalog]
$ cat zadanie1
root:x:0:0:root:/root:/usr/bin/zsh
```

Zad.4

a) `ls -l -p /etc | grep -v / | grep host`

```
(kali㉿kali)-[~]
$ ls -l -p /etc | grep -v / | grep host
-rw-r--r-- 1 root root 9 Aug 17 2025 host.conf
-r--r--r-- 1 root root 5 Sep 9 2025 hostname
-rw-r--r-- 1 root root 149 Dec 3 10:41 hosts
-rw-r--r-- 1 root root 411 Sep 9 2025 hosts.allow
-rw-r--r-- 1 root root 711 Sep 9 2025 hosts.deny
```

b) `awk -F : '{print $1}' /etc/passwd | sort`

```
(kali㉿kali)-[~]  
$ awk -F : '{print $1}' /etc/passwd | sort  
_apt  
avahi  
backup  
bin  
colord  
cups-pk-helper  
daemon  
Debian-snmpp  
dhcpcd  
dnsmasq  
games  
geoclue  
_gophish  
_gvm  
inetsim  
iodine  
irc  
kali  
lightdm  
list  
lp  
mail  
man  
messagebus  
miredo  
mosquitto  
mysql  
news  
nm-openconnect  
nm-openvpn  
nobody  
pipewire  
polkitd  
postgres  
proxy  
pulse  
redis  
redsocks  
root  
_rpc  
rtkit  
saned  
speech-dispatcher  
sshd  
sshd  
sshd  
statd  
strongswan  
stunnel4  
sync  
sys  
systemd-network  
systemd-timesync  
tcpdump  
tss
```

c) `awk -F : '{print $1}' | sort | tail -n 5`

```
(kali㉿kali)-[~]  
$ awk -F : '{print $1}' /etc/passwd | sort | tail -n 5  
tcpdump  
tss  
usbmux  
uucp  
www-data
```


d) `ping -c 5 127.0.0.1 | sed 's/ttl=64/ttl=128/'`

```
(kali㉿kali)-[~]
$ ping -c 5 127.0.0.1 | sed 's/ttl=64/ttl=128/'
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=128 time=0.059 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=128 time=0.062 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=128 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=128 time=0.077 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=128 time=0.065 ms

— 127.0.0.1 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4096ms
rtt min/avg/max/mdev = 0.059/0.065/0.077/0.006 ms
```

Zad.5

a) `ps -A | grep firefox`

```
(kali㉿kali)-[~]
$ ps -A | grep firefox
63914 ?        00:00:37 firefox-esr
```

b) `kill 63914` (id procesu firefoxa)

```
(kali㉿kali)-[~]
$ ps -A | grep firefox
63914 ?        00:00:38 firefox-esr

(kali㉿kali)-[~]
$ kill 63914
```

c) `ps -u kali` (kali to nazwa użytkownika)

```
(kali@kali)-[~]
$ ps -u kali
  PID TTY          TIME CMD
 1041 ?           00:00:01 systemd
 1043 ?           00:00:00 (sd-pam)
 1063 ?           00:00:05 dbus-daemon
 1064 ?           00:00:00 pipewire
 1065 ?           00:00:00 pipewire
 1067 ?           00:00:00 gnome-keyring-d
 1068 ?           00:00:00 mpris-proxy
 1069 ?           00:00:00 wireplumber
 1071 ?           00:00:00 pipewire-pulse
 1093 ?           00:00:13 xfce4-session
 1156 ?           00:00:00 VBoxClient
 1157 ?           00:00:10 VBoxClient
 1164 ?           00:00:00 VBoxClient
 1165 ?           00:01:08 VBoxClient
 1171 ?           00:00:00 VBoxClient
 1173 ?           00:03:27 VBoxClient
 1204 ?           00:00:00 at-spi-bus-laun
 1211 ?           00:00:01 dbus-daemon
 1221 ?           00:00:10 at-spi2-registr
 1231 ?           00:00:00 ssh-agent
 1235 ?           00:00:00 VBoxClient
 1237 ?           00:00:30 VBoxClient
 1250 ?           00:00:02 gpg-agent
 1253 ?           00:03:23 xfwm4
 1258 ?           00:00:00 gvfsd
 1264 ?           00:00:00 gvfsd-fuse
 1294 ?           00:00:16 xfsettingsd
 1300 ?           00:00:00 dconf-service
 1304 ?           00:00:20 xfce4-panel
 1310 ?           00:00:02 Thunar
 1317 ?           00:00:19 xfdesktop
 1319 ?           00:00:07 wrapper-2.0
 1327 ?           00:13:09 wrapper-2.0
 1328 ?           00:00:01 wrapper-2.0
 1329 ?           00:01:48 wrapper-2.0
 1330 ?           00:00:01 wrapper-2.0
 1331 ?           00:00:01 wrapper-2.0
 1332 ?           00:00:57 wrapper-2.0
 1342 ?           00:00:01 wrapper-2.0
 1369 ?           00:00:01 xiccd
 1370 ?           00:00:04 bluedman-applet
 1379 ?           00:00:00 agent
 1385 ?           00:00:12 xfce4-notifyd
 1393 ?           00:00:39 xfce4-screensav
 1402 ?           00:00:16 xfce4-power-man
 1422 ?           00:00:02 applet.py
 1424 ?           00:00:01 nm-applet
 1426 ?           00:00:01 polkit-mate-aut
 1458 ?           00:00:02 xcape
 1610 ?           00:00:00 obexd
 1611 ?           00:00:00 gvfs-udisks2-vo
 1639 ?           00:00:08 gvfs-afc-volume
```

Zad.dodatkowe

a) ps aux--sort-%mem | head -n 6

```
(kali@kali)-[~]
$ ps aux --sort=-%mem | head -n 6
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root        904   6.8   0.9 446132 161768 tty7      Ssl+  04:35   11:45 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -novtswitch
kali     1253   1.9   0.7 2075312 132752 ?        Sl     04:36   3:25 xfwm4
kali     1157   0.0   0.6 282620 109808 ?        Sl     04:36   0:10 /usr/bin/VBoxClient --clipboard
kali     1327   7.6   0.6 353540 105808 ?        Sl     04:36  13:10 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/plugins/libcpugraph.so 13 27262988 cp
the CPU load
kali     1317   0.1   0.5 529172 88524 ?        Sl     04:36   0:19 xfdesktop
```

- b) Komenda jobs pokazuje liste aktualnych zadań, bg przekazuje proces do działania w tle, fg mówi żeby się tym zajął na 'srodek ekranu' i nie można wtedy np. uzywac terminala bo jest zajęty przez ten proces

```
(kali㉿kali)-[~]
$ ps -A | grep firefox
72220 ?          00:00:10  firefox-esr

(kali㉿kali)-[~]
$ sleep 72220
^Z
zsh: suspended  sleep 72220

(kali㉿kali)-[~]
$ jobs
[1]  + suspended  sleep 72220

(kali㉿kali)-[~]
$ bg
[1]  + continued  sleep 72220

(kali㉿kali)-[~]
$ fg
[1]  + running    sleep 72220

bg
^Z
zsh: suspended  sleep 72220

(kali㉿kali)-[~]
$ fg
[1]  + continued  sleep 72220
█
```